



DF Metrics for Retail & Office Buildings Using Field DR Events Data

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July, 2022

Introduction

Background

Demand Flexibility Research

Building demand flexibility (DF) related research have been driven by the challenge of balancing electric supply and demand presented by the increasing penetration of renewable energy generation sources on the grid and the emergence of distribution energy resources (DERs).

DOE launched a Grid-Interactive Efficient Buildings (GEB) initiative that aims to optimize DERs for the benefit of building owners, occupants, and the grids.

DF from shed and shift can be triggered by grid dispatch or time-varying prices. There's a strong connection between DF and DR.

The Need for DF Benchmark

For grid operators and utilities, to understand the actual DF resource size from the building stock and what contributes to greater DF for a building, it's necessary to review field-measured DF performance from a wide range of buildings and compare.

The performance consistency and predictability are critical for grid planning and operations, which needs support from large dataset to understand performance variations.

For building managers, comparing the DF performance of a peer group can help identify best practices to be promoted and performance issues to be diagnosed.

DF Metrics in Literature

Existing DF metrics definitions and framework focus on prototype-based building energy simulation results and have not been applied to the actual performance of a significant group by real buildings to test their feasibility.

System-level

- ☐ System load impact
- ☐ System cost and emission impact

Building-level

- ☐ Theoretical characteristics of load shape changes
- ☐ Simulation-based building demand and comfort impact
- ☐ Simulation-based utility bill and comfort impact
- ☐ Field studies of DR performance

Objectives

❑ Propose and present a novel DF metrics framework for benchmarking.

- Understand the factors that influence real buildings' DF performance.
- Compare measured DF performance of buildings in different climates with evolving needs of the regional/local grid.

❑ Demonstrate applying the proposed framework tailored to stakeholders' needs.

- Office building dataset – understand the energy consumption implications of load shifting in real buildings.
- Retail building dataset (>100 buildings) - show example ways to tailor the benchmark in different dimensions.

Baseline

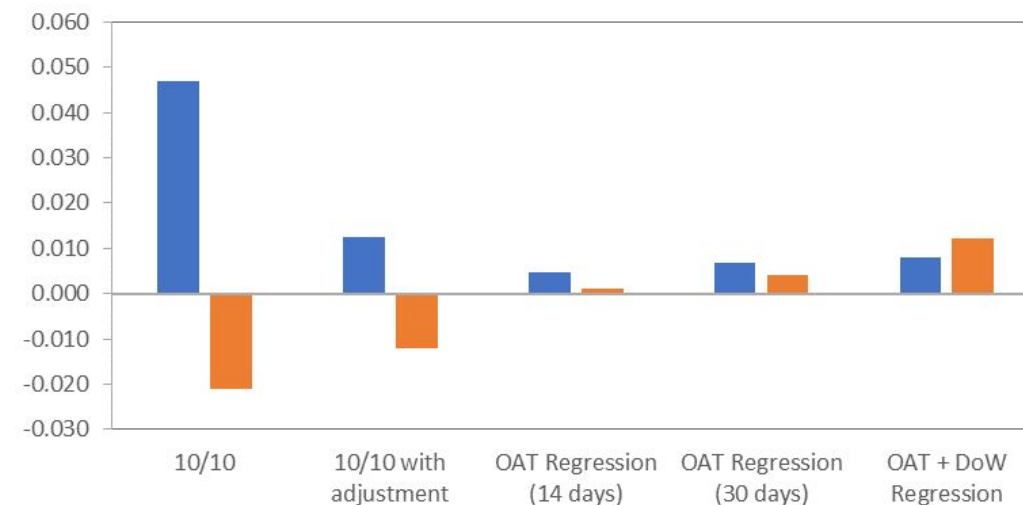
Baselines

Error metrics of baselines

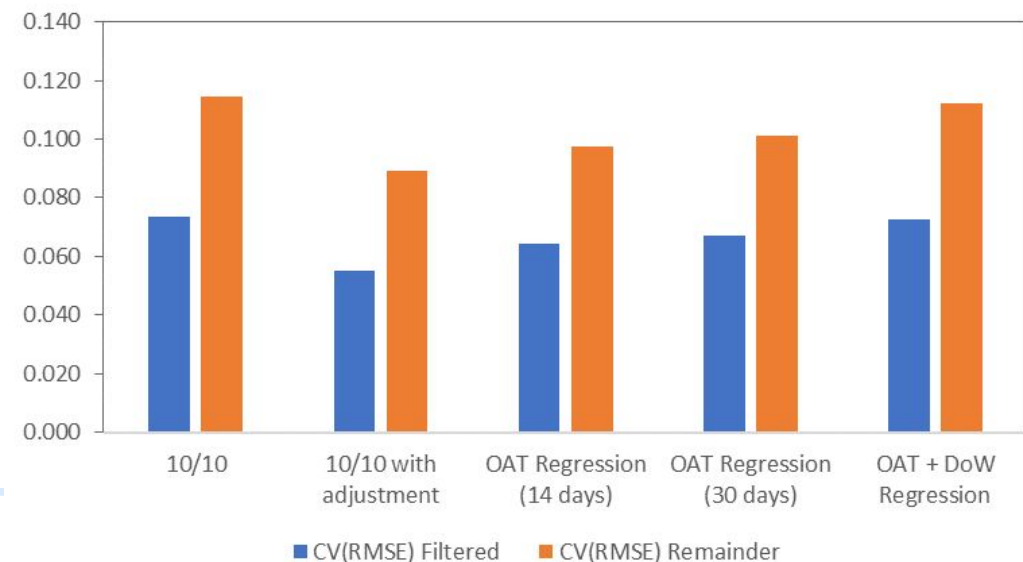
Model	nMBE		CV(RMSE)	
	Filtered	Remainder	Filtered	Remainder
10/10	0.047	-0.021	0.074	0.115
10/10 with adjustment	0.012	-0.012	0.055	0.089
OAT Regression (14 days)	0.005	0.001	0.064	0.098
OAT Regression (30 days)	0.007	0.004	0.067	0.101
OAT + DoW Regression	0.008	0.012	0.073	0.112

- ❑ All baselines are within ASHRAE Guideline 14 threshold.
- ❑ All baselines tend to under-predict the demand during “filtered” period when events are mostly called.
- ❑ “OAT Regression (14 days)” is considered to be the best.

nMBE



CV(RMSE)



■ CV(RMSE) Filtered ■ CV(RMSE) Remainder

Metrics Framework

Single-event Metrics

Single-event Metrics for Demand Decrease (“Shed”)

Metrics	Formula / Definition	Unit
E1: Net Building Consumption Change Percentage (24 hours)	= Net daily kWh change / Baseline daily kWh consumption x 100%	%
D1: Demand Decrease Intensity (DDI)	= D2 / Floor Area	W/ft ²
D2: Demand Decrease	= Average demand decrease during a single “shed” period	kW
D3: Demand Decrease Percentage (DDP)	= D2 / Baseline average demand during “shed” period	%

Single-event Metrics for Demand Increase (“Take”)

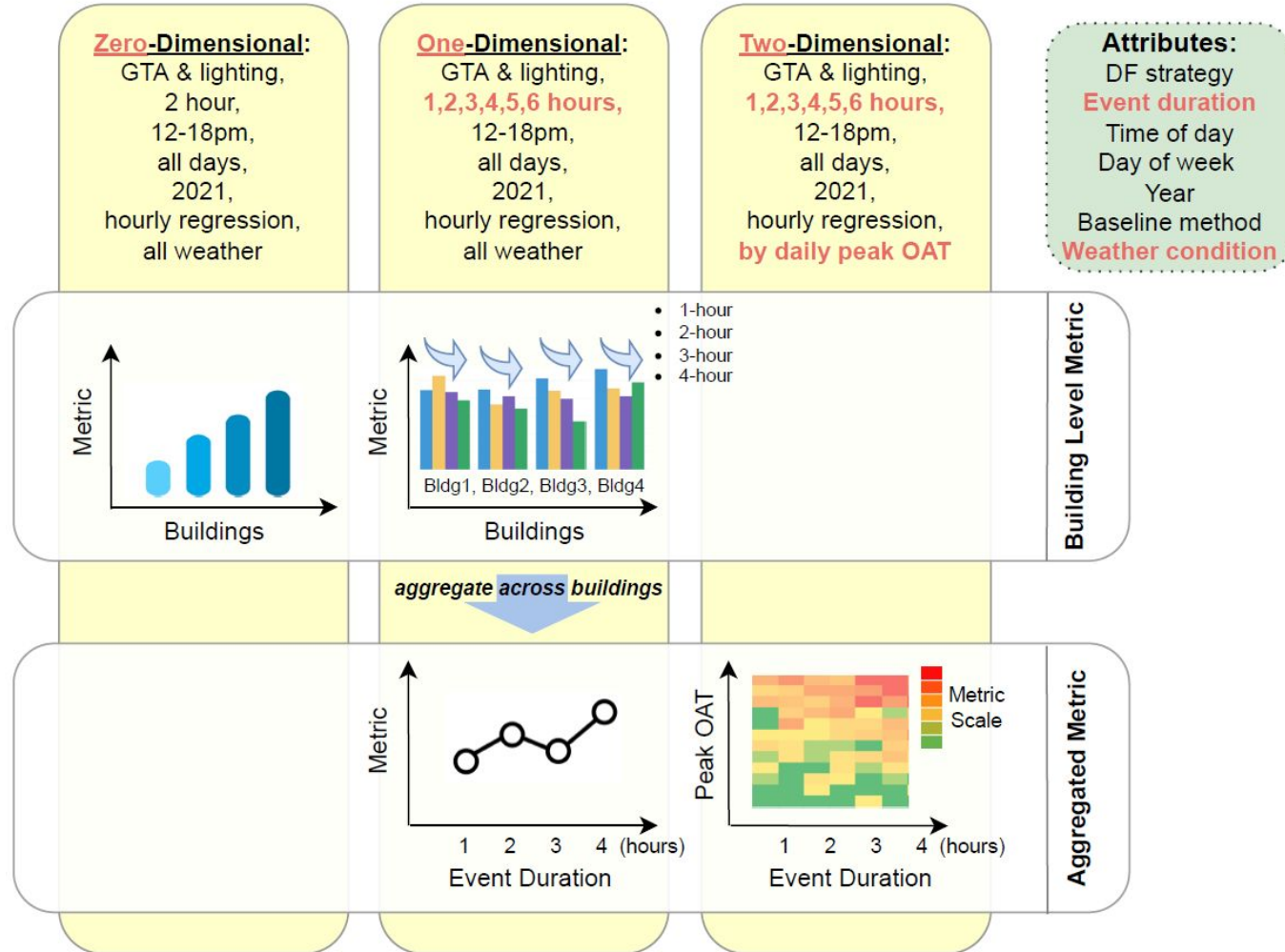
Metrics	Formula / Definition	Unit
I1: Demand Increase Intensity (DII)	= I2 / Floor area	W/ft ²
I2: Demand Increase	Average demand increase during a single “take” period	kW
I3: Demand Increase Percentage (DIP)	= I2 / Baseline average demand during “take” period	%

Benchmark Metrics

Benchmark Metrics for Shed and Shift

	Benchmarking Metrics	Formula / Definition	Unit
Peak, for reference	SPDI: Summer Peak Demand Intensity	= Summer hourly (or 15-min) peak demand / Floor area	W/ft ²
	WPDI: Winter Peak Demand Intensity	= Winter hourly (or 15-min) peak demand / Floor area	W/ft ²
Shed	Benchmark DDI (Benchmark Demand Decrease Intensity)	= Average of the five highest per-event $D1^2$ values ("top-5 shed")	W/ft ²
	Benchmark DDP (Benchmark Demand Decrease Percentage)	= Average of the per-event $D3^2$ values of the "top-5 shed"	%
	Benchmark DDS (Benchmark Demand Decrease Intensity Standard Deviation)	= Standard Deviation of the "top-5 shed"	W/ft ²
Take (Shift only)	Benchmark DI (Benchmark Demand Increase Intensity)	= Average of the $I1^3$ values associated with the "top-5 shed"	W/ft ²
	Benchmark DIP (Benchmark Demand Increase Percentage)	= Average of the $I3^3$ values associated with the "top-5 shed"	%
	Benchmark DIS (Benchmark Demand Increase Intensity Standard Deviation)	= Standard Deviation of the $I1^3$ values associated with "top-5 shed"	W/ft ²

Visualizing Benchmark Metrics

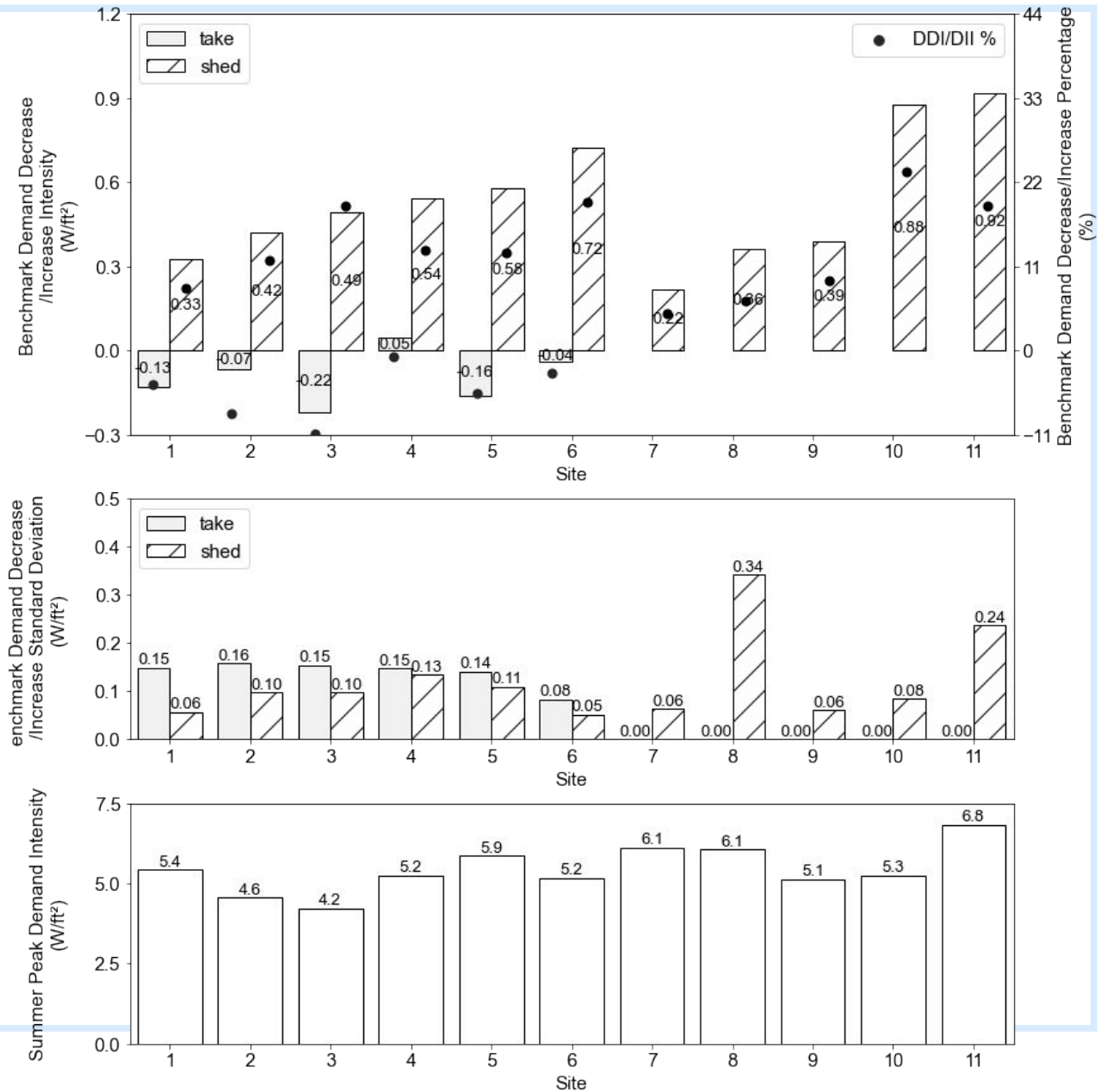


- ❑ **Zero-Dimensional:** All buildings in the group share the same values for all attributes
- ❑ **One-Dimensional:** One of the attributes has multiple values, and is of interest to understand its influence by minimizing influence from other attributes.
- ❑ **Two-Dimensional:** Two of the attributes have multiple values, and is of interest to understand their combined influence.

Office Buildings

Zero-Dimensional Benchmark

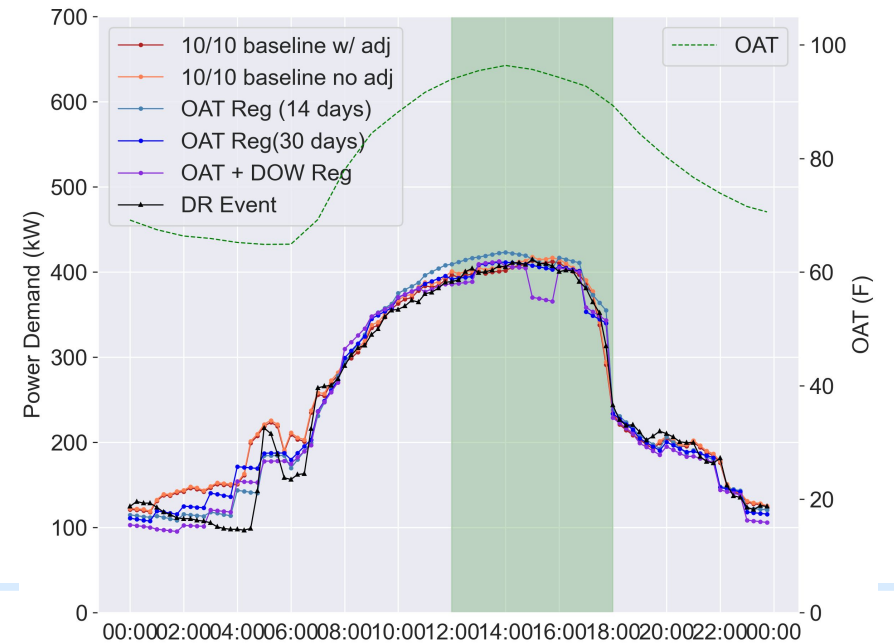
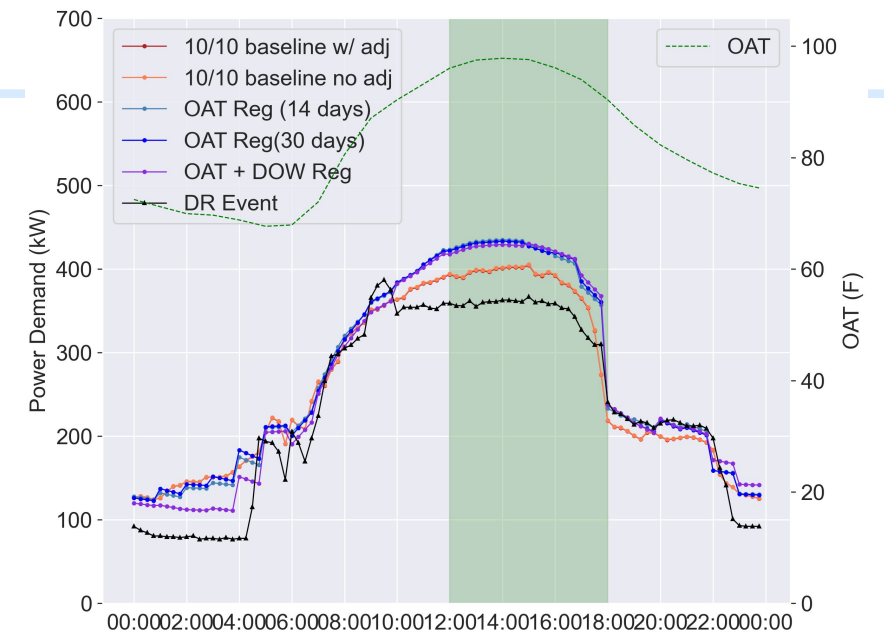
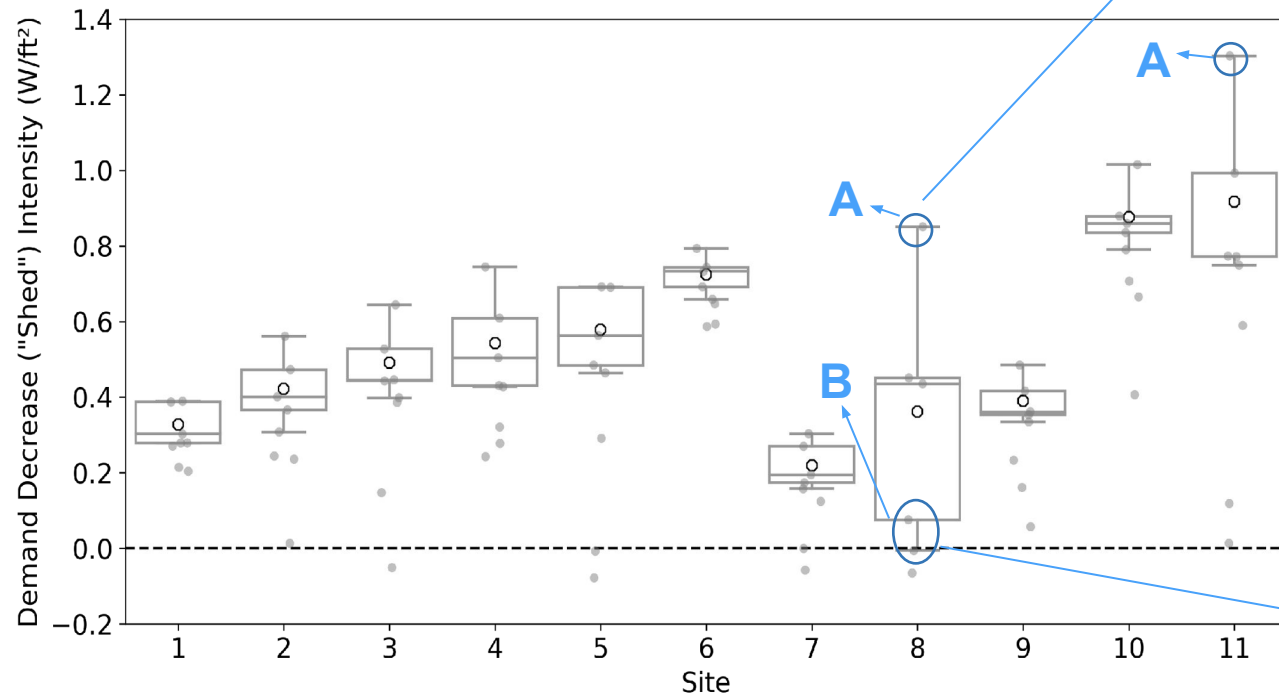
- 11 medium office buildings in southern California.
- HVAC: RTUs with VAV
- 2008 Data
- GTA (2-3 °F)
 - 5 of the buildings implemented pre-cooling.
- DR event window: 6-12 pm (6 hours)



Zero-Dimensional Benchmark

Reasons for wider boxes:

- A) The baseline is not predicting properly for some event days;
- B) No obvious shed observed on some event days.

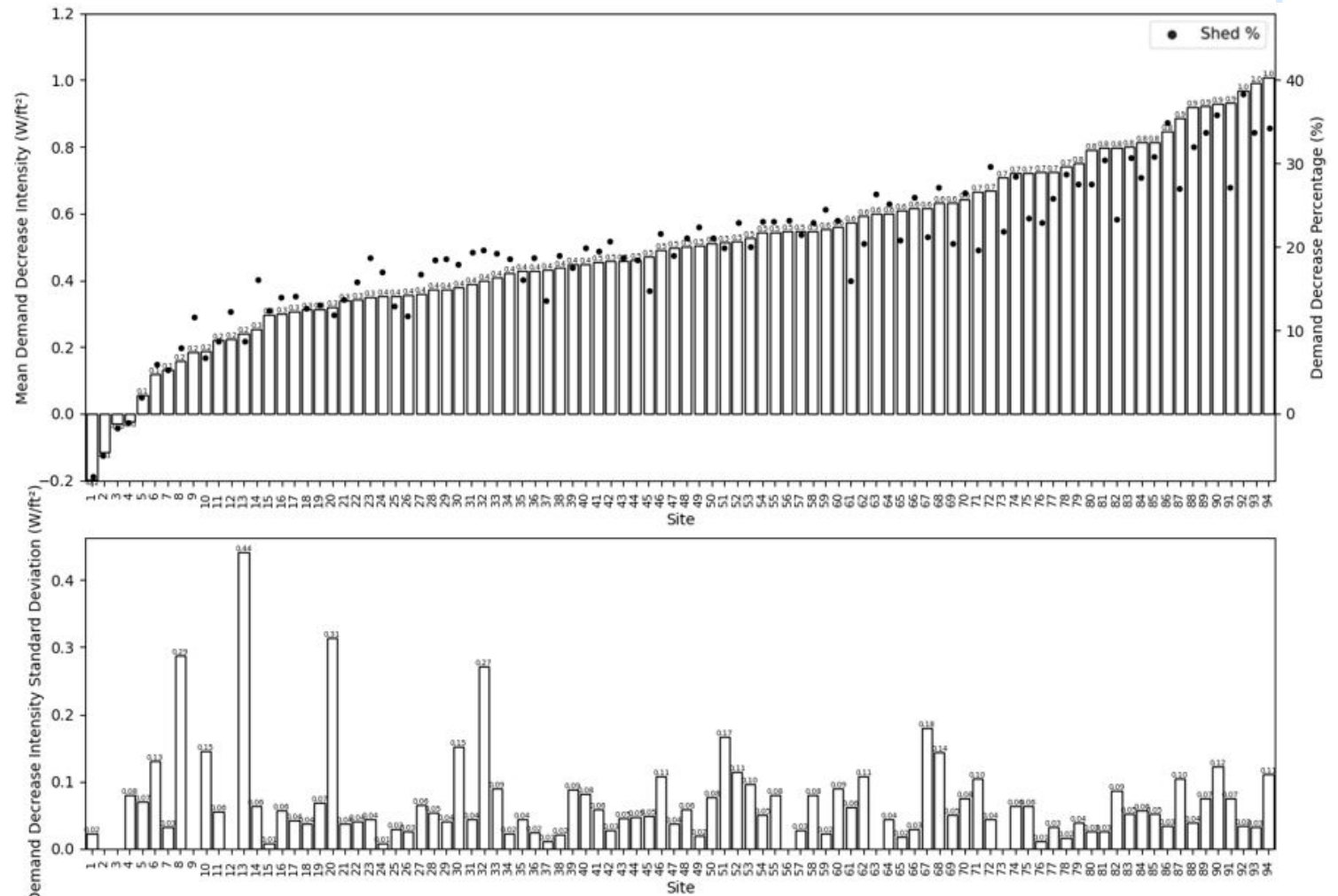


Retail Buildings

Zero-Dimensional Benchmark

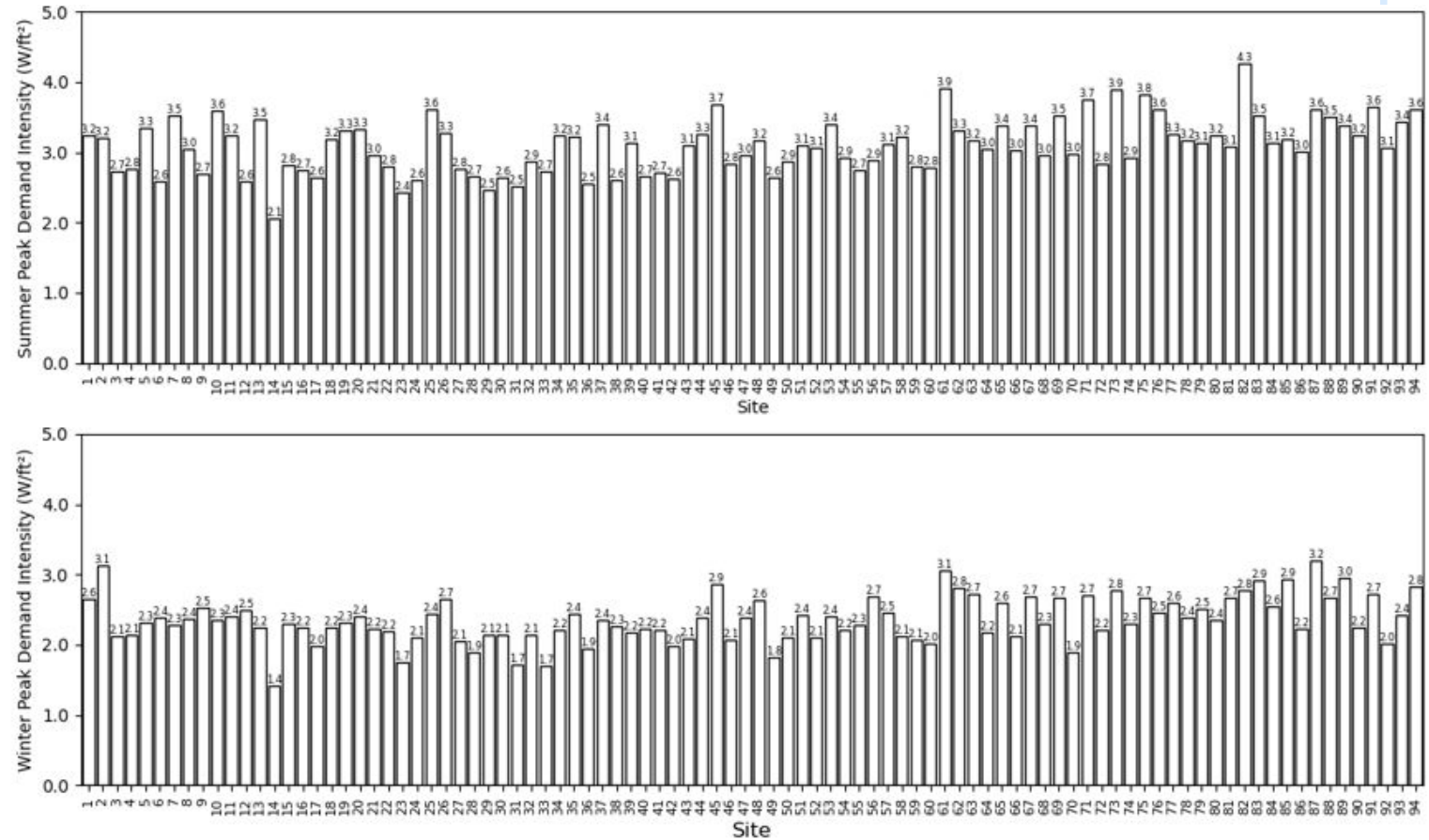
- ❑ 121 large big-box retail stores in 10 states. States that have the most sites include OH, NY, TX, and CA.
- ❑ HVAC: RTUs with variable speed fans.
- ❑ Shed strategies:
 - ~50% light dimming
 - Some of the RTUs deployed GTA
- ❑ DR event windows vary by site and by event. They are 1~6 hours long.

Benchmark DDI, DDP and DDS in **2021** for all sites (sorted by Benchmark DDI)

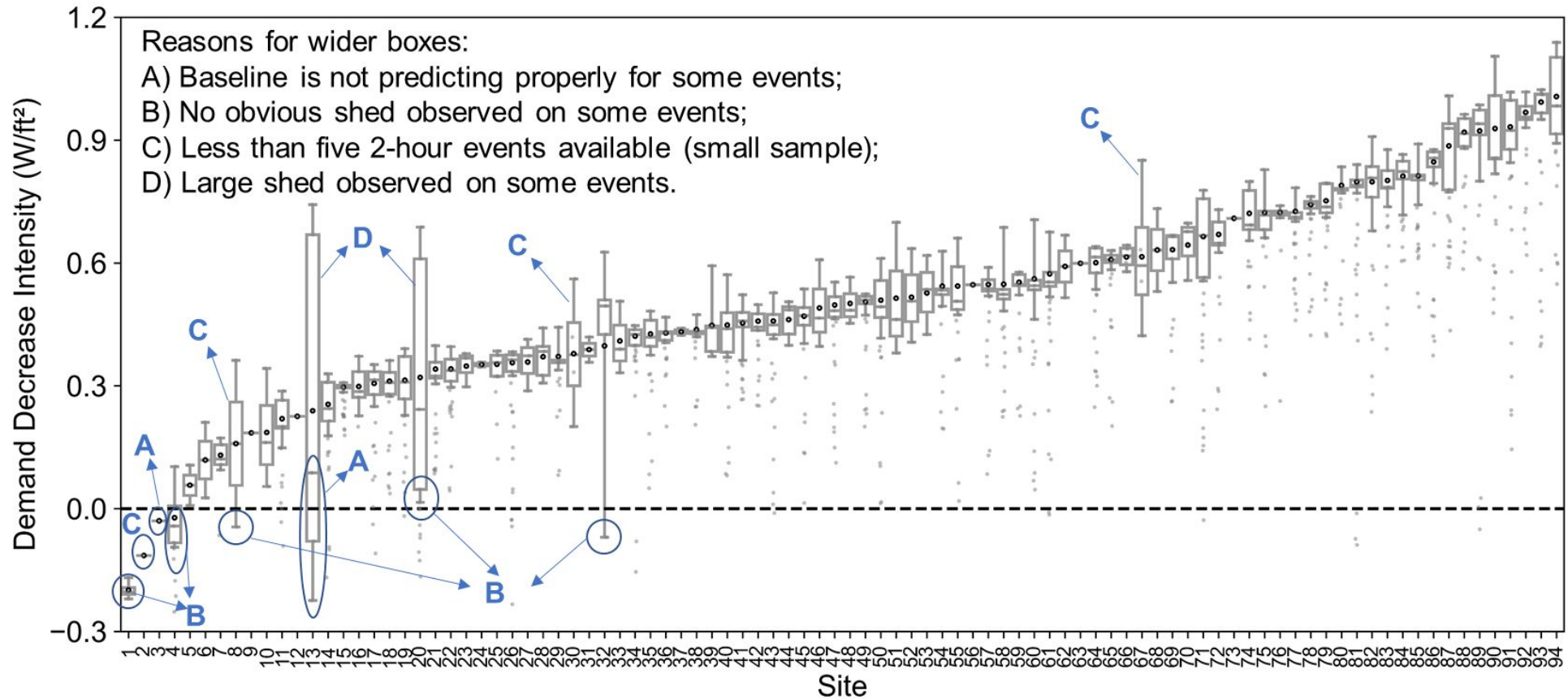


Zero-Dimensional Benchmark

Summer/Winter peaks in **2021** for all Batch 4 sites



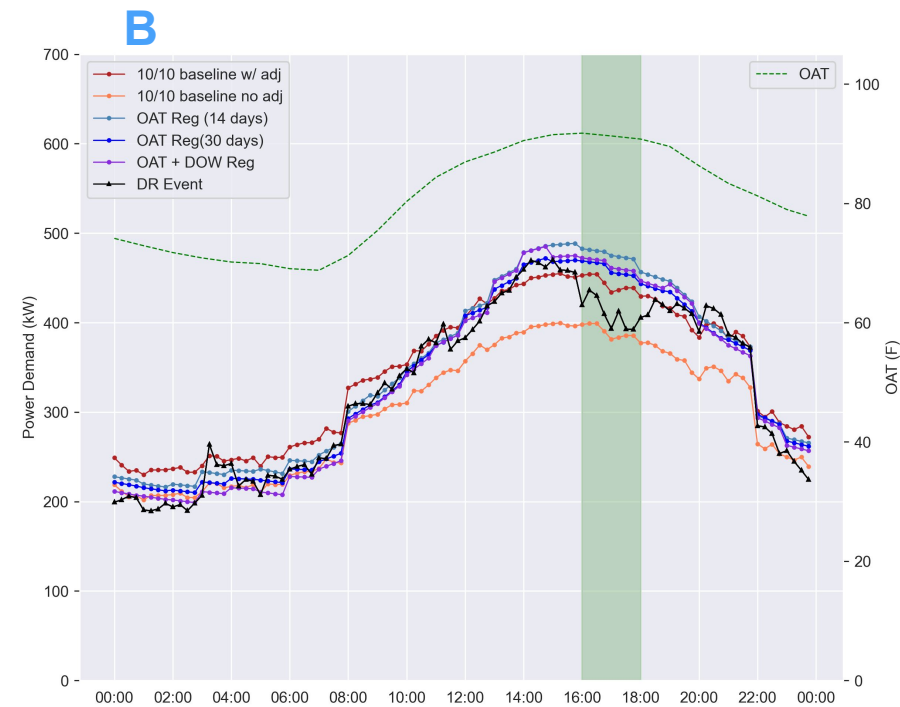
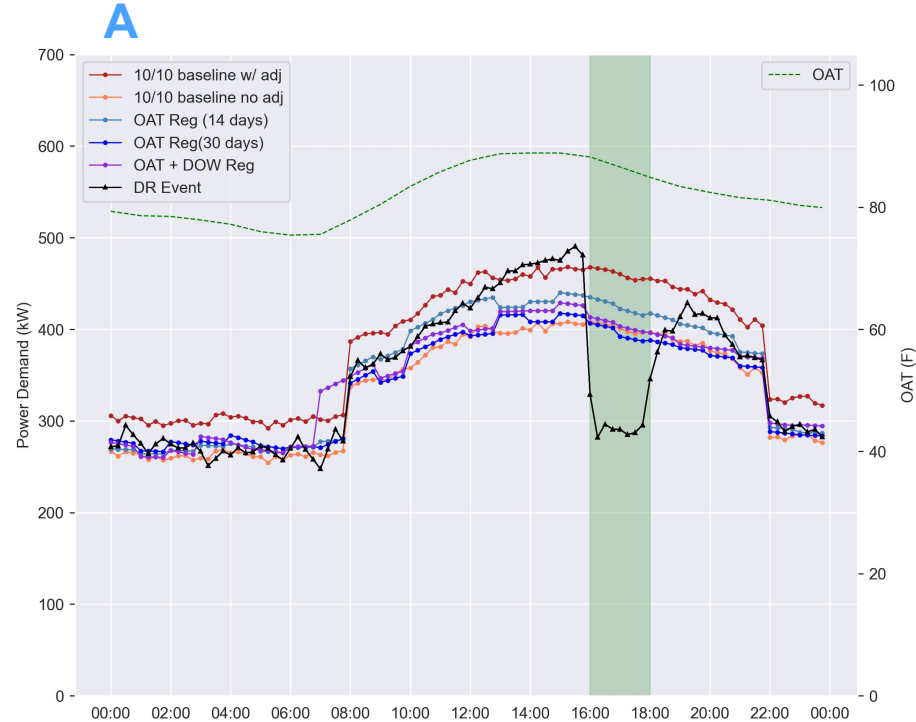
Zero-Dimensional Benchmark



Zero-Dimensional Benchmark

Reasons for wider boxes:

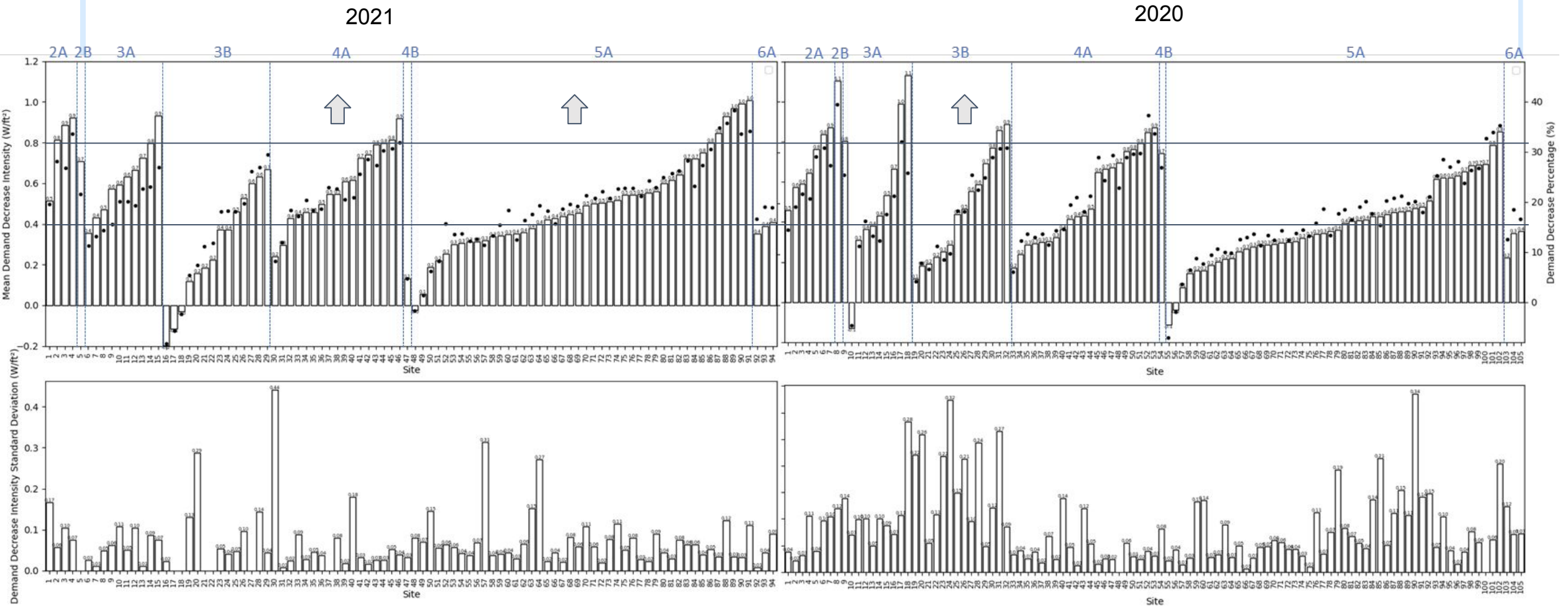
- A) Baseline is not predicting properly for some events;
- B) No obvious shed observed on some events;
- C) Less than five 2-hour events available (small sample);
- D) Large shed observed on some events.



Zero-Dimensional Benchmark

DDI Benchmark by Climate

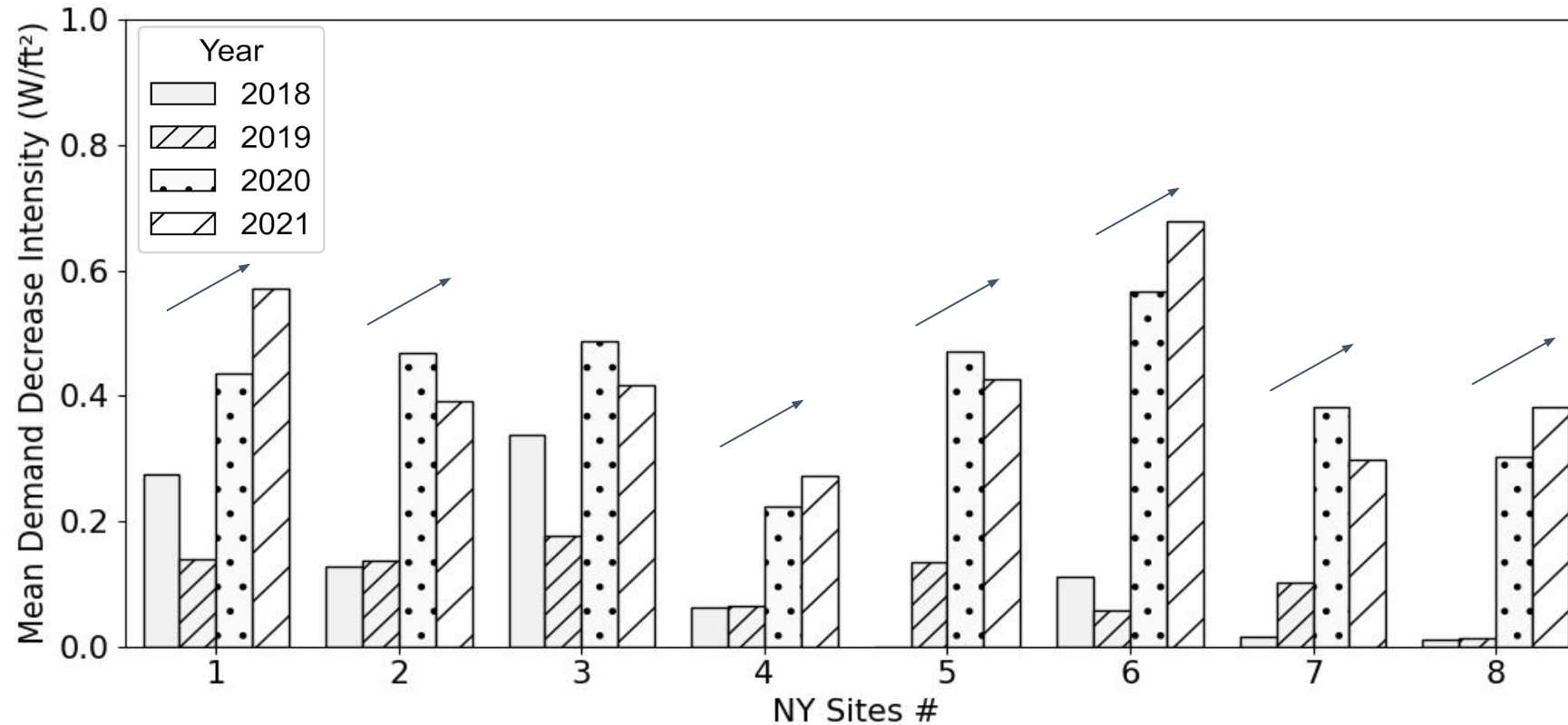
- Shed of 5A is lower than other climates in average.
- 2A and 3A are higher than other climates.



One-Dimensional Benchmark

Shed by Year

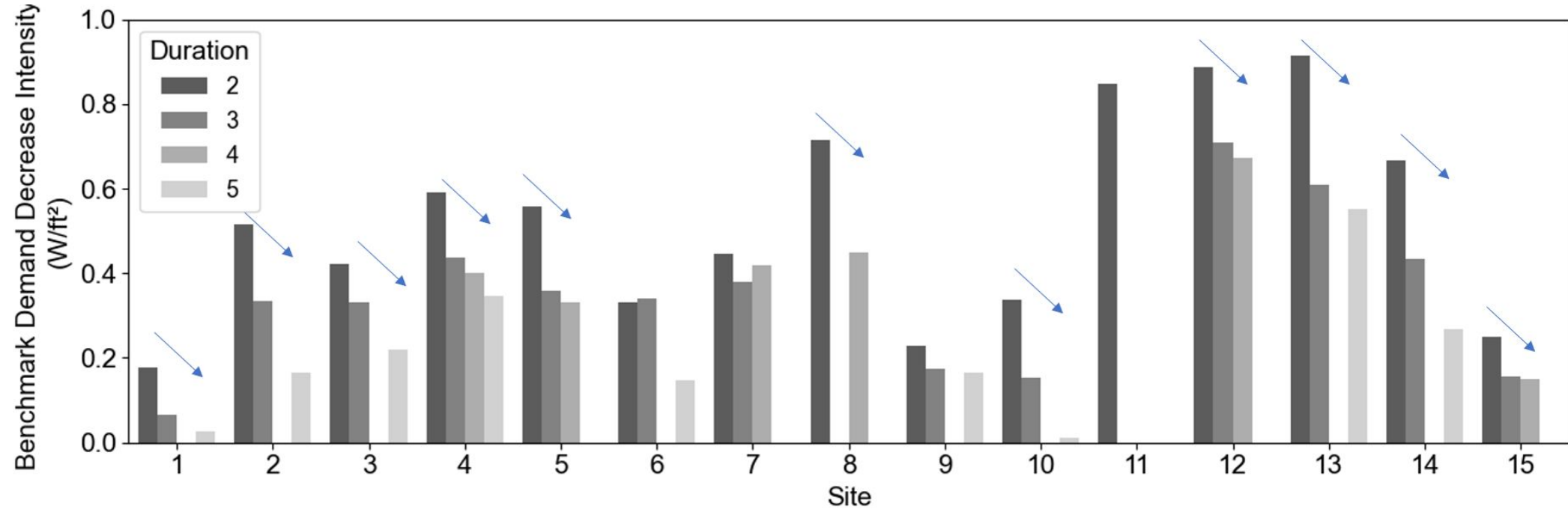
- The same building's DF performance can change significantly over time due to operational change, equipment retrofit or maintenance, etc.
- These stores upgraded their GTA algorithms between the DR seasons of 2019 and 2020.



One-Dimensional Benchmark

Most sites have a declining pattern of sheds by duration.

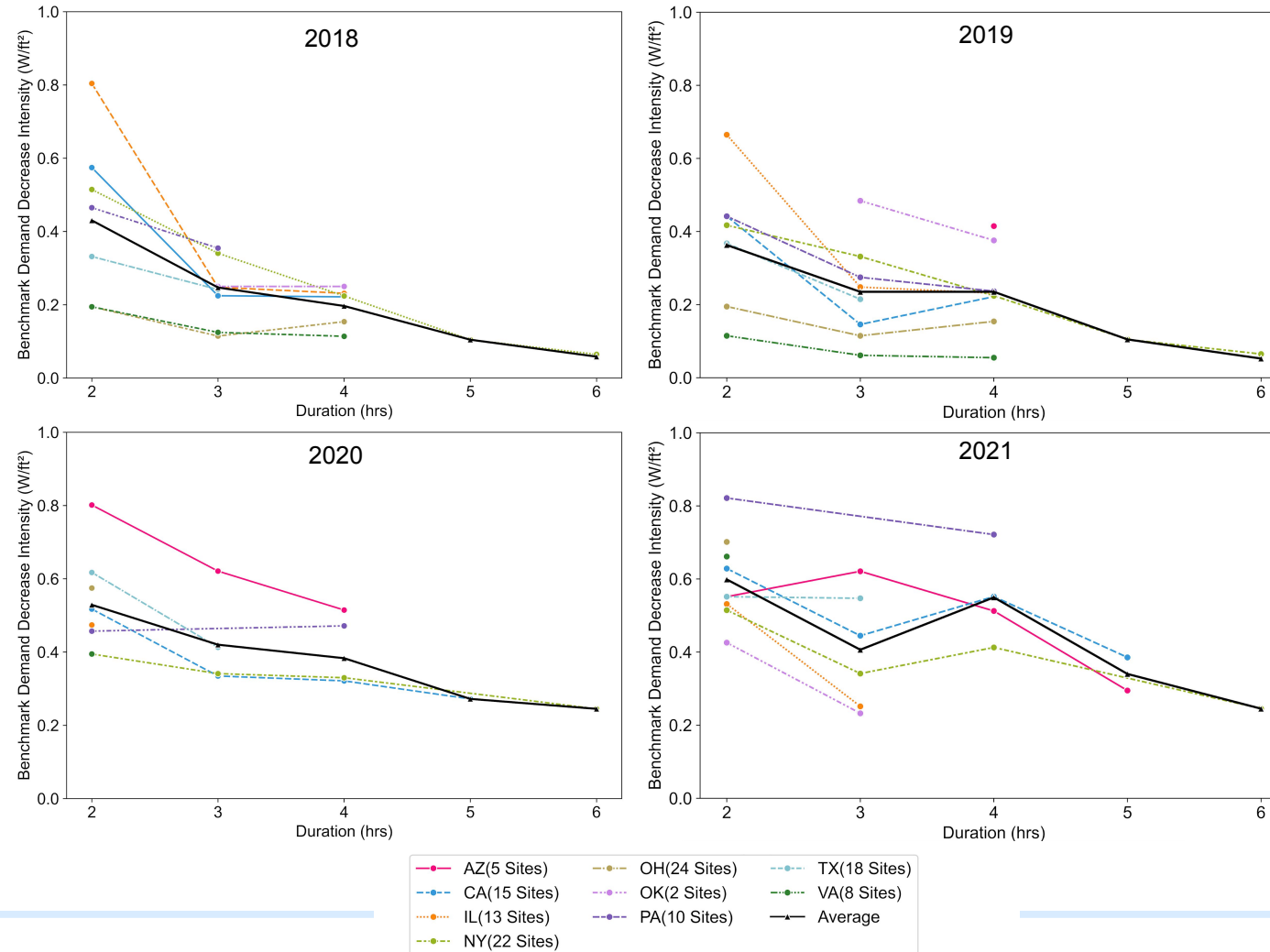
Shed vs Duration



One-Dimensional Benchmark

□ When aggregated to state-level, the overall average trend line also have a declining pattern.

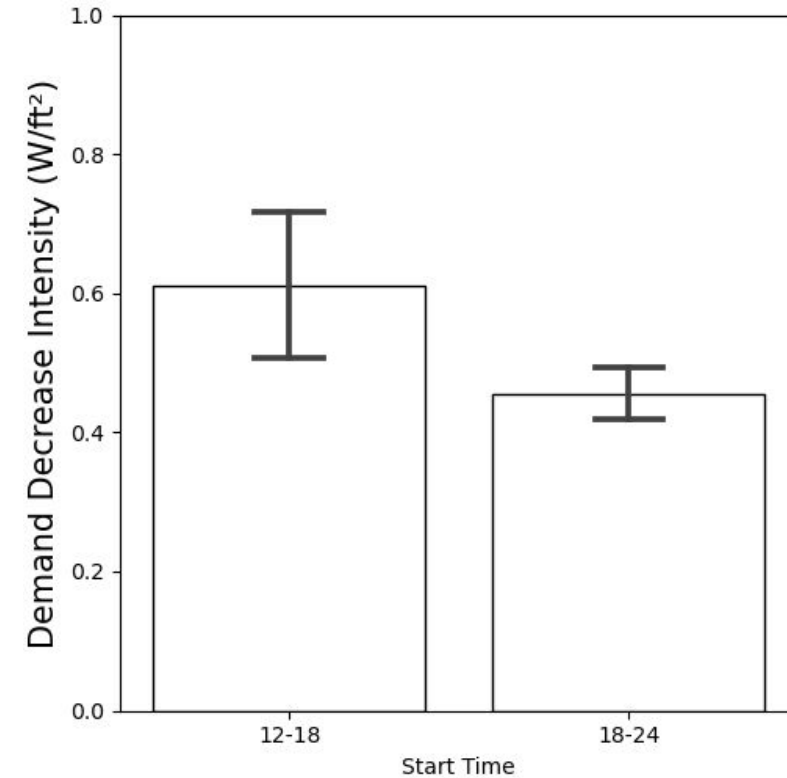
Shed vs Duration (By State)



One-Dimensional Benchmark

Shed vs Start Time

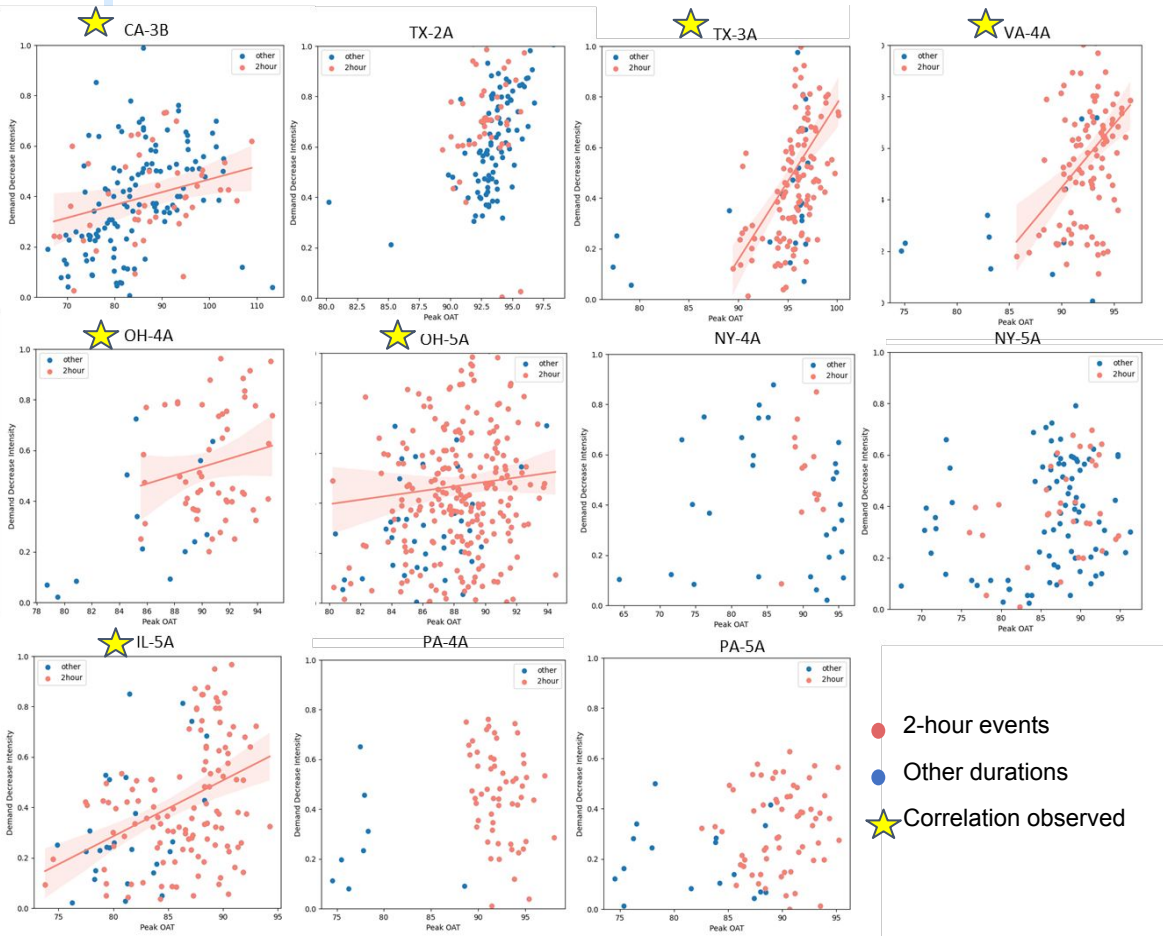
- ☐ *The benchmark DDI is higher for the afternoon events than the evening events.*
- ☐ *Lower cooling load and higher cooling efficiency during the cooler evening periods.*



One-Dimensional Benchmark

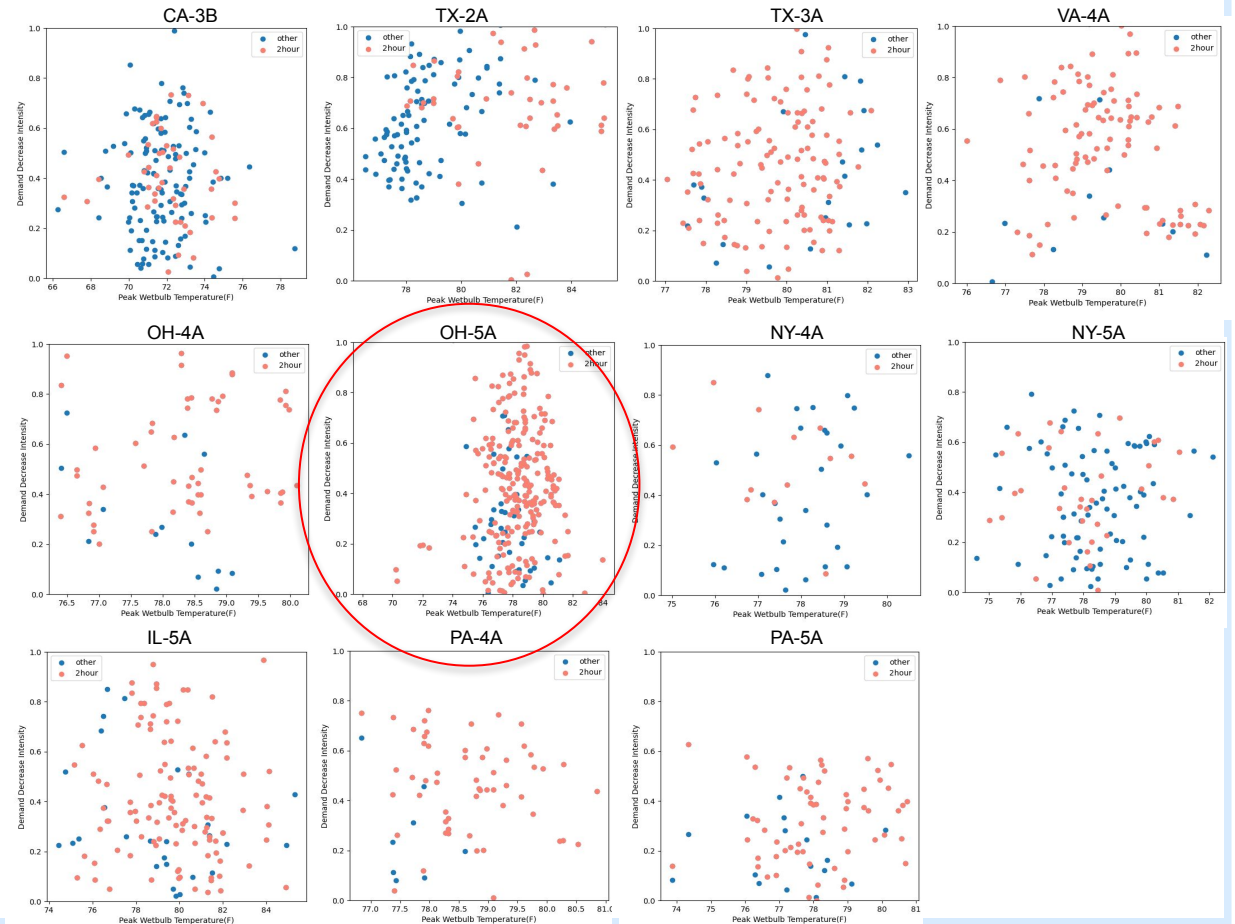
Shed vs OAT

2021



Shed vs Wetbulb

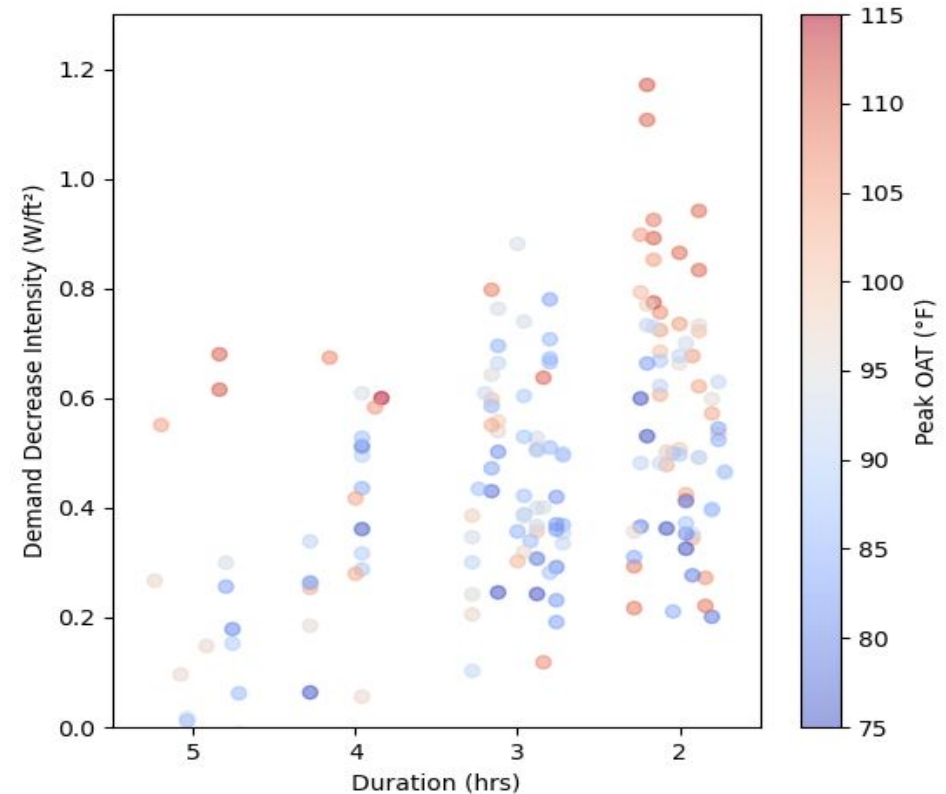
2021



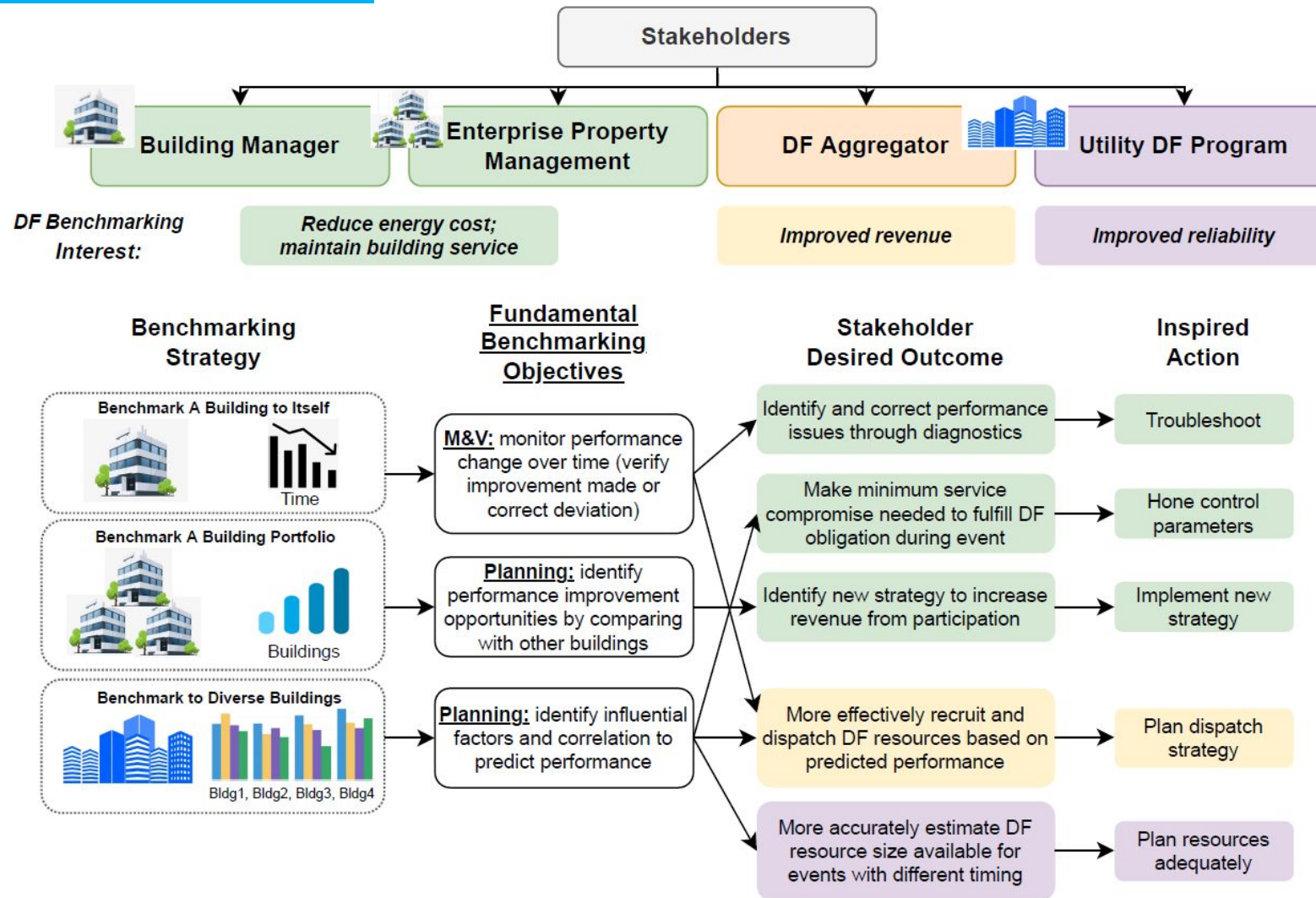
Two-Dimensional Benchmark

Shed vs Duration & OAT

- *In each duration group, the highest DDI occurs in higher temperature range.*
- *Highest DDI values in each duration group has a increasing trend as the duration gets shorter.*



Applications of DF Benchmark



Future Work

- ❑ Apply the framework to **large, diverse real building datasets** to accumulate empirical evidence-based understanding of commercial building DF.
- ❑ Implement this framework in a **public benchmarking database** with a **user interface** to enable various real-world applications.
- ❑ Expand this framework to **accommodate buildings with DERs** such as solar PV, battery storage, and EV.
- ❑ Link the DF metrics to **carbon metrics** to inform building decarbonization.
- ❑ Expand the framework to **address concerns on occupant comfort**.

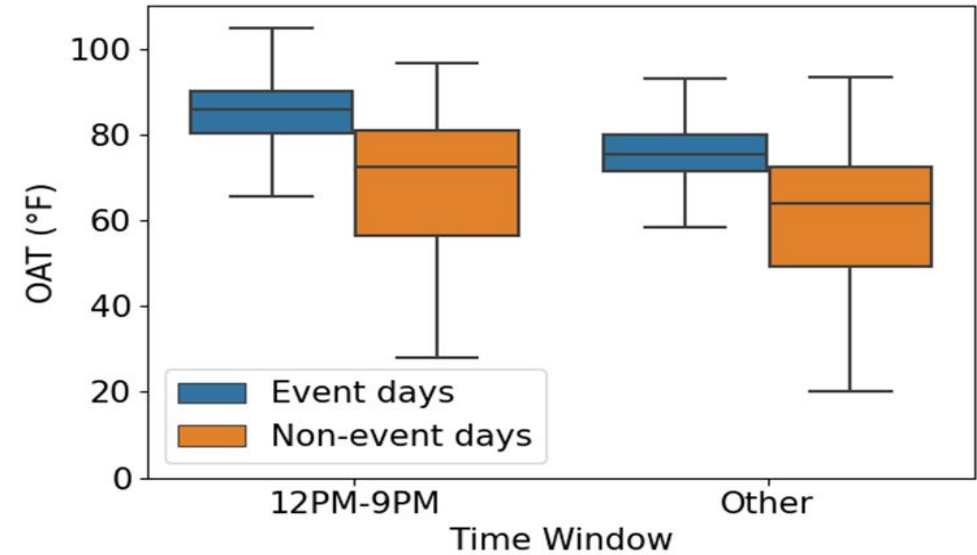
Appendix

Filtered: 12-9 pm, OAT > 80 °F

Error metrics of baselines

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Statistics on OAT on event days vs non-event days

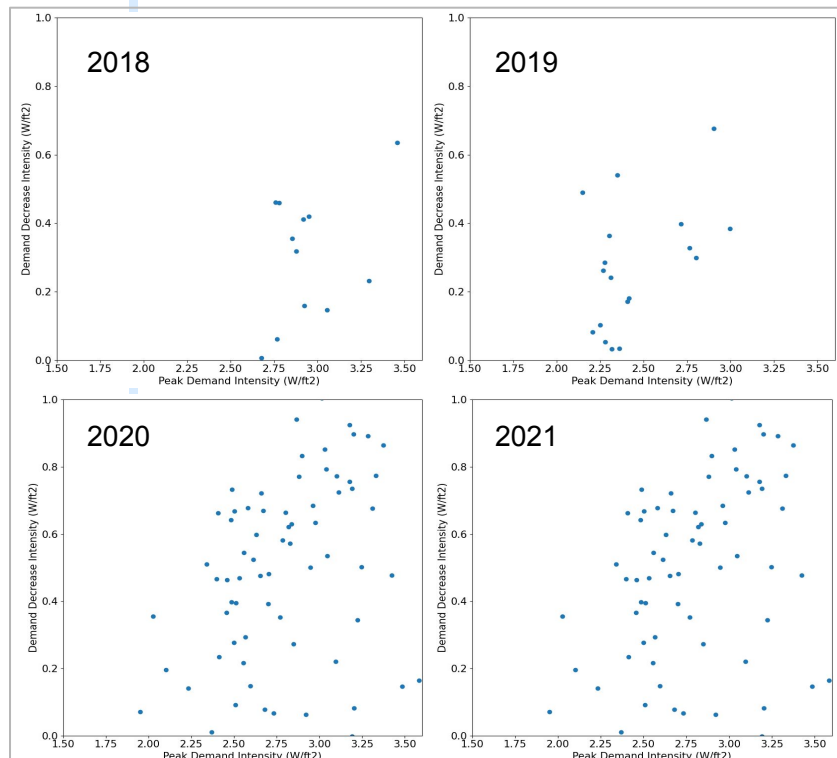


- ❑ Baselines are trained based on non-event days. The OAT is in generally lower than when events are called.

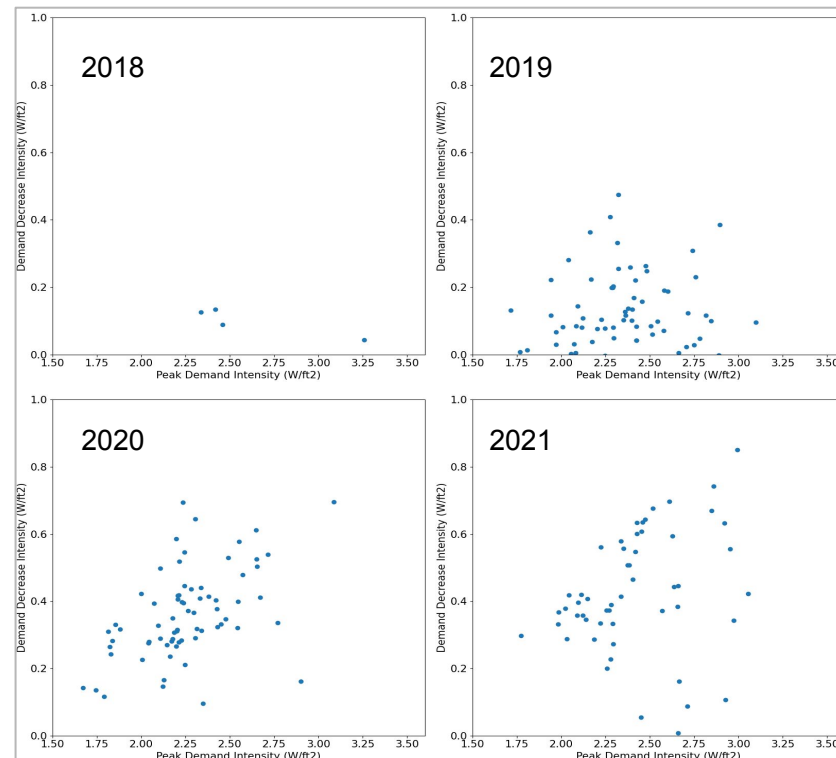
Appendix

Shed vs Peak

CA



NY



PA

